
Does AI-Assisted Writing Resurrect Retracted Science?

A Population-Level Study of Zombie Citations

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Abstract

Large language models trained before a retraction cannot know about it, and audits of chatbots show them drawing on retracted articles while flagging the retraction less than half the time. Whether that failure reaches the published literature is unknown. We measure it across 204.9 million citing works. Combining the Retraction Watch database with a frozen OpenAlex snapshot, we track citations to papers already retracted when they were cited, which we call zombie citations. Three designs answer the question in three ways, and their disagreement is the result. The population rate rises by a fifth at the release of ChatGPT, exceeding all 36 placebo breaks and concentrating in computational fields. Papers carrying phrases such as “as an AI language model” verbatim, where the filter reaches 70% precision, cite retracted work no more often than matched controls. A lexical AI score appears to separate the literature roughly hundredfold on the same outcome. That separation is article length: flagged papers carry 44.7 references against 9.2, and the proxy survives no reasonable specification once reference count is controlled. No per-paper design we can build attributes the population shift to AI assistance, and an excess-vocabulary proxy now used to compare LLM prevalence across fields and journals does not survive the obvious control. Data, code, and the artifact corpus are available at <https://github.com/garyzhang1006/ai-science-zombie-data>.

1 Introduction

Retraction is the scientific record’s error-correction mechanism, and it works only if authors check the current status of what they cite. Large language models complicate that mechanism in a specific, foreseeable way: a model whose training data was collected before a retraction has no representation of it, and a model asked to summarize a field will reproduce the retracted claim with the same fluency as any other. The chatbot-level evidence is already troubling, since a recent audit found that ChatGPT drew on 61.5% of retracted stem-cell articles when answering domain questions and mentioned the retraction in only 41.7% of those responses (Zhang et al., 2025), and a broader study reached similar conclusions across fields (Thelwall et al., 2025). What no study has measured is whether the published literature itself now cites retracted work at an elevated rate because of AI-assisted writing. That is the question this paper answers.

The measurement problem is harder than it looks, and the difficulty shapes our design. Citations to retracted papers are rare, AI involvement is observed only through noisy proxies, and a naive regression of zombie-citation rates on a lexical AI score is attenuated by proxy error, so a null from that regression would describe the proxy rather than the literature. We therefore built the study around three prongs whose failure modes do not overlap. The first is a population interrupted time

series, which needs no per-paper attribution at all. The second is a cohort in which AI involvement is near-certain, because the papers carry verbatim LLM artifacts such as “as an AI language model” that usually reach print only through unedited pasting. The third makes the lexical score of Kobak et al. (2024) and Liang et al. (2025) honest instead of discarding it, using partial identification under misclassification (Manski, 2003).

Running all three was worth the cost, because they disagree. The population series moves at the release of ChatGPT, while the cohort with certain exposure does not move at all, and the lexical AI score separates sharply until reference count is controlled and then barely at all. A study built on any one prong would have reported a clean result and been wrong about what it measured.

In the same cohorts we track a second, denser outcome: references that do not resolve, which prior evidence links to model fabrication far above the human baseline (Walters and Wilder, 2023; Bhattacharyya et al., 2023). A zombie citation is real-but-refuted knowledge propagating; an unsolvable reference is fabricated support.

We contribute four things. The first is a population-level measurement of zombie-citation dynamics across the LLM transition, covering 204.9 million citing works and 216,514 zombie citation events. The second is the artifact cohort itself, a public corpus of 1,731 papers with certain LLM involvement and matched controls, with a measured false-positive rate for every phrase we use. The third is a set of bounded rather than attenuated estimates of the dose-response between lexical AI involvement and citation integrity. The fourth is a warning: that proxy’s apparent association with citation integrity is almost entirely reference count, and what remains does not hold across specifications.

2 Related work

Post-retraction citation is a longstanding integrity concern predating language models, covering the persistence of citations to retracted work and interventions such as the RISRS recommendations (Schneider et al., 2022). Case studies have traced the citation network of a single prominent retracted paper and found direct citation to be the main propagation channel (van der Vet and Nijveen, 2016). We add scale and a treatment contrast, measuring the phenomenon across the full indexed literature and asking whether the LLM transition changed its rate.

On the AI side, audits of conversational models document the retraction blindness described above (Zhang et al., 2025; Thelwall et al., 2025). They also document fabricated references. Walters and Wilder measured 55% of GPT-3.5 citations and 18% of GPT-4 citations as fabricated, with substantive errors in a further 24% of the GPT-4 citations that do point at real work (Walters and Wilder, 2023; Bhattacharyya et al., 2023). Strzelecki catalogued verbatim artifact phrases across premier journals (Strzelecki, 2025), establishing that unedited model output reaches print; we industrialize that into a matched-cohort design. Prevalence estimation (Kobak et al., 2024; Liang et al., 2025; Thelwall, 2025) supplies our dose proxy. To our knowledge no prior work connects these threads to measured citation-integrity outcomes.

3 Data and design

3.1 Retraction and citation data

Retraction events come from the Retraction Watch database, which yields 60,247 retractions carrying both a usable DOI and a retraction date, of which 59,924 (99.5%) match a work in OpenAlex. Citation events and reference lists come from the OpenAlex parquet snapshot of 26 June 2026. We pin that single release so that every number here is reproducible against a frozen corpus rather than a moving index. We work from the snapshot rather than the REST API by necessity: the API now meters usage at 1,000 calls or \$0.10 per day on its free tier, which cannot cover a scan of half a billion works, and the snapshot is unmetered.

A citation is a zombie citation if the citing paper appeared at least six months after the cited paper’s retraction date, the buffer absorbing indexing lag. We exclude citing works that mention retraction in their title or abstract, since citing a retracted paper to discuss its retraction is the correction system working as intended. Scanning all 2,446 snapshot files yields 216,514 zombie citation events across 204.9 million citing works published between 2018 and 2026. Both the numerator and the denominator come from that same pass, which is why the monthly rate needs no sampling correction.

Table 1: Artifact phrase registry with false-positive rates measured against the pre-ChatGPT era. A phrase appearing in a paper published before 2023 cannot be unedited model output, so the pre-2023 share of hits estimates the phrase’s false-positive rate.

Phrase	Hits	Pre-2023	FP rate	Status
as an AI language model	9,131	111	1.2%	retained
as a large language model, I	244	1	0.4%	retained
as of my last knowledge update	157	3	1.9%	retained
as an AI developed by OpenAI	99	3	3.0%	retained
my knowledge cutoff	89	0	0.0%	retained
I don’t have access to real-time	22	1	4.5%	retained
I cannot browse the internet	3	0	0.0%	retained
Regenerate response	9,085	7,926	87.2%	excluded

87 3.2 The artifact cohort

88 We harvest candidate artifact papers by full-text search over a registry of verbatim LLM phrases.
89 This step needs more care than it first appears, and getting it wrong would have destroyed the cohort.
90 OpenAlex full-text search is stemmed, so the phrase “regenerate response” matches “regenerative
91 response” and returns 9,085 hits dominated by zebrafish and stem-cell regeneration papers. Those
92 hits would have made up 82% of our cohort.

93 We caught this with a negative control that the study period supplies for free. No paper published
94 before 2023 can contain unedited ChatGPT output, so every pre-2023 hit is a false positive by con-
95 struction, and the pre-2023 share of a phrase’s hits estimates its false-positive rate directly. Table 1
96 reports that rate for every phrase we screened. “Regenerate response” fails at 87.2% and is excluded.
97 The seven surviving phrases range from 0.0% to 4.5%, with the workhorse phrase “as an AI lan-
98 guage model” at 1.2% across 9,131 hits. This converts the claim that these phrases make model
99 involvement certain from an assertion into a measurement.

100 Filtering removes papers whose own subject is language models, since those quote the phrases
101 deliberately: 3,613 candidates are dropped on title, 1,868 on subfield, and 1,549 for carrying no
102 references. The filter is imperfect. On a seeded 100-paper sample of the retained cohort its precision
103 is 70% (61 to 79), the misses being AI-topic papers where the phrase may be quotation. What
104 remains is 1,731 papers, all published in 2023 or later, of which 1,643 carry a venue identifier. We
105 match each to five controls from the same venue and year, within 25% of its reference-list length
106 and nearest in citation count, giving 7,644 matched pairs. Reference-list lengths come out nearly
107 identical across arms, 42.55 against 42.45, so the matching binds. Within this cohort the exposure
108 is close to observed rather than proxied, at the price of representing only the least careful tail of
109 AI-assisted writing, a selection we return to in Section 6.

110 3.3 Outcomes, series, and bounds

111 For every paper we compute two outcomes over its reference list: the count of zombie citations
112 as defined above, and the count of unresolvable references, meaning entries whose DOI fails to
113 resolve in Crossref and whose bibliographic string matches no indexed work under fuzzy search
114 at a similarity threshold of 85. The population series aggregates the first outcome monthly within
115 field strata, and the interrupted time series allows a level and slope change at the ChatGPT release
116 in December 2022, with Newey-West standard errors and placebo breaks at each month of 2019
117 through 2021 calibrating the procedure’s own false-positive rate.

118 The bounding analysis follows Manski (2003). Given a lexical AI score with assumed sensitivity
119 and specificity, the observed outcome-by-score table set-identifies the true outcome rates among AI-
120 assisted and unassisted papers. We solve this over a grid rather than at any single assumed point,
121 and the data disciplines the grid. Our rule flags 0.86% of 2023-onward papers, and the prevalence
122 solution requires specificity above $1 - P(S=1) = 0.991$. Any specificity below that admits more
123 false positives than observed positives, and so no solution at all. A rule demanding two distinct
124 markers from a 37-term list is high-specificity and low-sensitivity by construction, and the grid sits
125 in that regime.

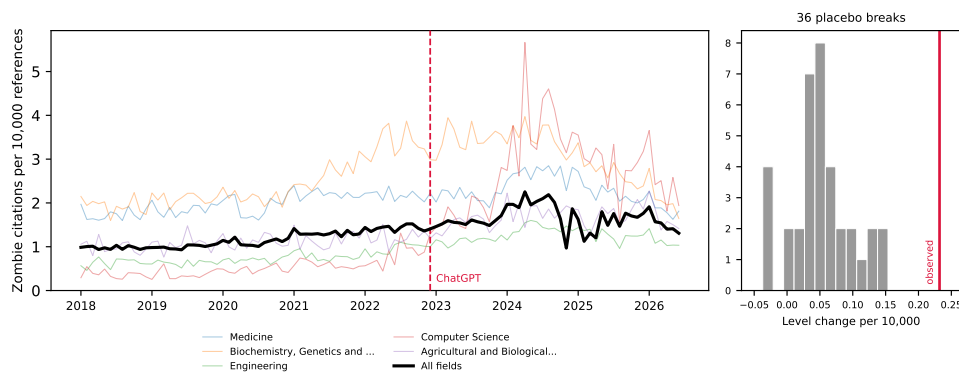


Figure 1: Zombie citations per 10,000 references, monthly, 2018–2026, for the five fields contributing the most events and for the literature as a whole. The vertical line marks the ChatGPT release; the inset shows the distribution of 36 placebo break estimates from 2019–2021 against the estimated 2022 break.

126 The outcome for this prong is a per-paper indicator, whether a paper carries at least one zombie
 127 citation, and that indicator rises with the length of a paper’s reference list. The lexical rule needs two
 128 markers inside an abstract, so it selects papers with long abstracts, which are full research articles
 129 carrying many references. Across the unrestricted sample, flagged papers average 44.7 references
 130 against 9.2 for the rest. We therefore compute the bounds within reference-count strata, and take
 131 papers with at least 20 references as the primary specification, which compares full articles to full
 132 articles at 61.2 references against 50.8.

133 4 Results

134 We take the three prongs in order of how much they assume. The population series assumes nothing
 135 about individual papers, the artifact cohort observes exposure directly, and the lexical AI score infers
 136 it.

137 4.1 The population rate shifts at the ChatGPT release

138 The population rate of zombie citation runs at 1.160 per 10,000 references over the pre-period and
 139 rises through it, and the interrupted time series estimates a level change of $+0.232$ at the ChatGPT
 140 release (standard error 0.130, $p = 0.074$), a 20% increase on the pre-period rate. The conventional
 141 p -value is marginal, but it is the weaker of the two tests available here. Across 36 placebo breaks
 142 placed through 2019 to 2021, the largest absolute level change is 0.154 and their standard deviation
 143 is 0.047, so the real break sits five placebo standard deviations out and exceeds every placebo the
 144 procedure produces on data where nothing happened, a placebo-based p below $1/37$ (Figure 1).
 145 Placebo fits use only pre-break months, so they run on shorter series than the real fit, which widens
 146 their spread and makes the comparison conservative.

147 The level change is stable under truncation: refitting with the series ending in June 2025, December
 148 2025, March 2026, and June 2026 gives $+0.193$, $+0.190$, $+0.194$, and $+0.232$. The slope change
 149 is not. It reads -0.0116 ($p = 0.005$) on the full series but decays to -0.0086 ($p = 0.312$) when the
 150 last year is dropped, and the snapshot’s final months are visibly under-indexed, carrying 9.0 million
 151 references in June 2026 against a monthly norm near 15 million. We therefore report the level shift
 152 as the population result and treat the post-break decline as an artifact of the indexing tail.

153 The break concentrates where one would expect AI writing tools to have landed first. Computer
 154 Science shows a level change of $+1.489$ ($p = 0.006$). Biochemistry follows at $+0.606$ ($p = 0.043$),
 155 then Engineering at $+0.296$ ($p = 0.010$), Agricultural and Biological Sciences at $+0.297$ ($p =$
 156 0.005), and Materials Science at $+0.174$ ($p = 0.017$). Medicine, the largest field by volume, does
 157 not move ($+0.163$, $p = 0.362$). Eight fields were tested, and Computer Science ($p = 0.0057$) and
 158 Agricultural and Biological Sciences ($p = 0.0045$) are the two that survive a Bonferroni threshold

Table 2: Artifact cohort against matched controls. Rates per 1,000 references; ratios with 95% cluster-bootstrap intervals resampling papers. The unresolvable-reference row rests on 7,873 papers and 415,750 references checked against Crossref.

Outcome	Artifact papers	Matched controls	Ratio
Zombie citations	0.157	0.179	0.878 [0.379, 1.632]
Unresolvable references	138.84	127.74	1.087 [1.005, 1.174]
Reference-list length	42.55	42.45	

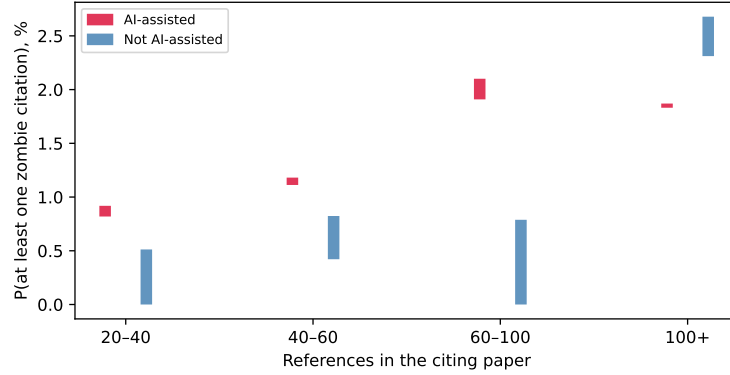


Figure 2: Set-identified probability of carrying at least one zombie citation, by lexical AI score, within strata of reference-list length. Bars span the full range of solutions across the grid. Each flagged bar rests on 6 to 11 events, so the strata illustrate the direction of the adjustment rather than establishing a dose-response.

159 of 0.00625; we read the rest as suggestive of a computational concentration rather than as separately
160 established.

161 4.2 Near-certain LLM involvement predicts nothing

162 Within the cohort where LLM involvement is near-certain, the zombie-citation result is a null. Arti-
163 fact papers cite retracted work at 0.157 per 1,000 references against 0.179 among controls, a ratio of
164 0.878 whose 95% interval, 0.379 to 1.632, comfortably contains parity (Table 2). This is a bounded
165 null rather than an absence of evidence: the design rules out effects above roughly 1.6 times the
166 control rate, and the point estimate sits below one. The endpoint is simply sparse. At 0.157 per
167 1,000 references, 1,645 artifact papers carrying 42.55 references each are expected to produce about
168 eleven zombie citations in total, and no cohort of this size can resolve a modest effect on such a rate.

169 The denser endpoint carries what statistical weight the cohort has, as the design anticipated. Arti-
170 fact papers carry 138.84 unresolvable references per 1,000 against 127.74 among controls, a ratio of
171 1.087 with an interval of 1.005 to 1.174. The effect is real but small, and its interval touches parity
172 closely enough that we decline to describe it as robust. Its absolute level also deserves care: an unre-
173 solvable rate near 13% across both arms is far too high to represent fabrication, and mostly reflects
174 Crossref indexing gaps for books, theses, and items without DOIs. Only the ratio between matched
175 arms is interpretable, and only under the assumption that those indexing gaps fall symmetrically
176 across artifact and control papers drawn from the same venue and year.

177 4.3 The lexical AI score measures article length

178 Applied without restriction to a seeded sample of 629,900 papers from 2023 onward, this prong
179 gives the study’s largest contrast and a spurious one. The rule flags 0.86% of papers. Among those,
180 1.057% carry at least one zombie citation against 0.113% among the rest, and the correction set-
181 identifies [1.24%, 5.11%] against [0.00%, 0.089%], intervals that never overlap on the grid. The
182 comparison is about article type. Flagged papers average 44.7 references against 9.2, because a rule

183 needing two markers inside an abstract selects long-abstract full articles, and the outcome counts
184 whether a paper carries any zombie citation at all.

185 Restricting to papers with at least 20 references brings the arms to 61.2 against 50.8, and the naive
186 contrast falls from roughly ninefold to 1.55 (1.261% against 0.816%). Stratifying leaves flagged
187 papers higher in three of four strata and lower in the fourth (Figure 2), but those cells rest on 6 to 11
188 flagged events, and the reversal above 100 references is 6 events against 105 at a Fisher p of 0.84.
189 We report the strata for the shrinkage across them and decline to read a dose-response into them.

190 The test that settles the question fits the event indicator on the proxy and on log reference count
191 together, across all 78,465 papers with at least 20 references and 652 events. Reference count
192 dominates, at an odds ratio of 2.21 per log unit with $p = 2 \times 10^{-38}$, against 1.35 for the proxy
193 with a 95% interval of 0.95 to 1.92 and $p = 0.097$. On the full sample the proxy appears significant
194 (odds ratio 1.54, $p = 0.008$) only because reference count there ranges from 1 to several hundred
195 and carries an odds ratio of 3.71.

196 Sweeping both thresholds shows how little the proxy contributes. Across twelve specifications,
197 crossing marker thresholds of one to three with reference cuts at 10, 20, 30, and 40, the adjusted odds
198 ratio stays between 1.20 and 1.74 and eight intervals contain one. The four that reach significance
199 are those with the loosest reference cut, where the most short papers remain and residual length
200 confounding is largest. We read this as a small association that no specification test establishes,
201 which is far from the hundredfold separation the unconditioned fit reports.

202 This is the paper’s most transferable result. Excess-vocabulary proxies already compare prevalence
203 across subpopulations: Kobak et al. (2024) report a lower bound that “differed across disciplines,
204 countries, and journals, reaching 40% for some subcorpora.” Any such comparison inherits whatever
205 else the markers track, and pairing those estimates with an outcome inherits it twice. Misclassification
206 bounds do not protect against this. The correction assumes the proxy errs at random with respect
207 to the outcome, and length-driven selection violates that in the direction that inflates the result.

208 5 Discussion

209 The three prongs give three different answers, and the disagreement is the paper’s substance. The
210 population rate moves at the ChatGPT release by an amount that survives every placebo the data can
211 generate, and the move concentrates in the fields where AI writing tools arrived first. The cohort
212 with near-certain exposure shows nothing at all: its zombie endpoint sits below parity, and its denser
213 endpoint clears parity by a margin a reviewer may call fragile. The lexical AI score seems to split
214 the literature, then holds no specification-robust signal once reference count enters the model.

215 The three now agree on the per-paper question, since neither observed nor inferred exposure pre-
216 dicta citing retracted work, and the design that appeared to disagree was measuring article length
217 rather than writing practice. That is the failure mode the bounds machinery was built to expose and
218 could not, because misclassification correction assumes the proxy errs at random with respect to the
219 outcome. The cohort where exposure is observed is where a causal effect should be largest, and it
220 is absent there. What the proxy separates is a population differing from the rest of the literature in
221 more ways than its use of language models, most obviously in how much it cites.

222 This leaves the population result standing and unexplained, which we think is the honest place to
223 leave it, given that the break concentrates in computational fields, survives 36 placebo breaks and
224 four series endpoints, and coincides with the arrival of AI writing tools. It is equally consistent
225 with contemporaneous changes in indexing, in citation practice, or in who published in those fields
226 in 2023. Separating these needs per-paper exposure that is neither self-reported nor lexical, a gap
227 public data cannot close.

228 The cohort null needs one qualification, since it is selected on carelessness and therefore bounds
229 the careless tail rather than AI assistance at large. That tail is nonetheless where integrity screening
230 would start, which makes the null useful to anyone planning such a programme.

231 For the workshop’s framing question, the study offers a measured instance of a general worry and
232 a sharper caution about how the worry gets measured. The aggregate rate at which the literature
233 cites retracted work did shift when AI writing tools arrived, but whether AI assistance caused that

234 shift remains unresolved by our strongest design, and the excess-vocabulary proxies now in wide
235 circulation would have answered the question confidently and, on this evidence, wrongly.

236 6 Limitations

237 The unresolvable-reference comparison queried every one of the 9,009 intended papers, of which
238 7,873 (87.3%) deposit reference data in Crossref at all; the remainder are absent from the comparison
239 for a reason unrelated to their content. OpenAlex full-text coverage is partial and skewed toward
240 open access, so the artifact cohort undercounts paywalled carelessness. The artifact filter’s precision
241 is 70% (61 to 79) on a seeded 100-paper sample, adjudicated by a language model on title and
242 abstract rather than by a human on full text. Using a model to audit a model-artifact filter is not
243 independent, and we report the figure as indicative. The era test bounds how often a phrase fires on
244 text no model could have written. It says nothing about a 2023-or-later author quoting the phrase
245 while discussing model output, which the topic and title filters target instead. We publish the full
246 candidate list with every drop reason, together with a seeded 100-paper sample, so that any reader
247 can audit both. Retraction Watch coverage is itself uneven across fields and eras. More seriously,
248 the interrupted time series identifies a break at the ChatGPT release but cannot by itself attribute it
249 to LLM writing rather than to contemporaneous changes in indexing or citation practice, which the
250 placebo distribution mitigates without eliminating. The bounds correct for misclassification of the
251 lexical AI score and not for confounding, and Section 4 shows that the confounding is large enough
252 to dominate the unconditioned estimate; the conditioned estimate holds reference count fixed but
253 nothing else, so field, venue, and author language remain uncontrolled. Finally, all three prongs
254 measure citation to retracted work, and none measures whether the citing text endorses the retracted
255 claim, so our rates are upper bounds on endorsement.

256 7 Conclusion

257 The literature’s zombie-citation rate rose about 20% when AI writing tools arrived, against a pre-
258 period rate of 1.160 per 10,000 references, and the rise concentrates in computational fields. The
259 papers we can show a language model wrote display no such elevation, which leaves the population
260 result standing without a mechanism we can name, and we would rather report that gap than close it
261 by picking whichever prong tells the cleaner story. The measurement lesson outlasts this particular
262 question, because an excess-vocabulary proxy of the kind now used to compare LLM prevalence
263 across disciplines separated the literature roughly hundredfold on citation integrity. None of that
264 separation survived a control for how many references a paper carries. The artifact cohort with its
265 measured per-phrase false-positive rates, the bounds machinery, and the monthly series are released
266 at <https://github.com/garyzhang1006/ai-science-zombie-data>.

267 References

- 268 X. Zhang, T. Gu, S. Loganathan, Y. Xie, and J. Chen. Retracted articles on stem cells are contin-
269 uously cited in publications and used by ChatGPT. *Scientometrics*, 130(11):6503–6511, 2025.
270 doi:10.1007/s11192-025-05484-y
- 271 M. Thelwall, M. Lehtisaari, I. Katsirea, K. Holmberg, and E. Zheng. Does ChatGPT ig-
272 nore article retractions and other reliability concerns? *Learned Publishing*, 38(4), 2025.
273 doi:10.1002/leap.2018
- 274 A. Strzelecki. ‘As of my last knowledge update’: How is content generated by ChatGPT in-
275 filtrating scientific papers published in premier journals? *Learned Publishing*, 38(1), 2025.
276 doi:10.1002/leap.1650
- 277 D. Kobak, R. González-Márquez, E. Á. Horvát, and J. Lause. Delving into LLM-assisted writing in
278 biomedical publications through excess vocabulary. *arXiv:2406.07016*, 2024.
- 279 W. Liang, Y. Zhang, Z. Wu, et al. Quantifying large language model usage in scientific papers.
280 *Nature Human Behaviour*, 9(12):2599–2609, 2025. doi:10.1038/s41562-025-02273-8
- 281 M. Thelwall. ChatGPT contamination: estimating the prevalence of LLMs in the scholarly literature.
282 *arXiv:2403.16887*, 2025.

- 283 W. H. Walters and E. I. Wilder. Fabrication and errors in the bibliographic citations generated by
284 ChatGPT. *Scientific Reports*, 13(1):14045, 2023. doi:10.1038/s41598-023-41032-5
- 285 M. Bhattacharyya, V. M. Miller, D. Bhattacharyya, and L. E. Miller. High rates of fab-
286 ricated and inaccurate references in ChatGPT-generated medical content. *Cureus*, 2023.
287 doi:10.7759/cureus.39238
- 288 C. F. Manski. *Partial Identification of Probability Distributions*. Springer, 2003.
- 289 J. Schneider, N. Woods, R. Proescholdt, et al. Reducing the inadvertent spread of retracted science:
290 recommendations from the RISRS report. *Research Integrity and Peer Review*, 7(1):6, 2022.
291 doi:10.1186/s41073-022-00125-x
- 292 P. E. van der Vet and H. Nijveen. Propagation of errors in citation networks: a study involving the
293 entire citation network of a widely cited paper published in, and later retracted from, the journal
294 Nature. *Research Integrity and Peer Review*, 1(1):3, 2016. doi:10.1186/s41073-016-0008-5

295 A Artifact phrase registry and filtering

296 Table 1 gives the full registry with raw counts and measured false-positive rates. The filter drops
297 candidates in a fixed order and records a reason for each. Phrase retired for stem collision accounts
298 for 9,088, published before 2023 for 119, and subject classified in an AI subfield for 1,868. A
299 title matching a language-model topic pattern drops 3,613, an abstract carrying two or more meta-
300 discussion terms drops 363, a document type outside the article, review, chapter, preprint, and dis-
301 sertation set drops 531, and an empty reference list drops 1,549. These counts sum to the 18,862
302 unique candidates; 1,731 survive. The complete candidate table with per-paper decisions ships in
303 the repository as `artifact_candidates.csv`, and a seeded 100-paper sample for hand validation
304 ships as `hand_validation_sample.csv`.

305 B Robustness and practical considerations

306 Section 4 refits the series at four endpoints, where the level change varies between +0.190 and
307 +0.232 while the slope change loses significance. The six-month buffer, the 25% matching toler-
308 ance, and the fuzzy-match threshold of 85 live in `config.py`; `bounds.json` reports the unrestricted
309 fit, the primary fit, and every stratum.

310 Four decisions in this pipeline each cost a day and changed a number in this paper. We record them
311 because the errors are specific to citation-scale measurement.

312 **Stemmed full-text search collides with domain vocabulary.** Our registry originally included “re-
313 generate response,” which OpenAlex stems and matches against “regenerative response.” It returned
314 9,085 hits, 87.2% of them predating ChatGPT, drawn from zebrafish and stem-cell regeneration
315 work, and it was 82% of the cohort before removal. The era test in Table 1 caught it, and every
316 phrase now passes that test before admission.

317 **OpenAlex serves abstracts as an inverted index, not as text.** Abstracts arrive as JSON mapping
318 each word to its positions, so scoring the serialized string matched no markers: they appear as quoted
319 dictionary keys. The failure is silent, returning a proxy-positive rate of zero and unidentified bounds
320 rather than an error. Parsing the index first gives 0.86%.

321 **A per-paper outcome indicator inherits reference-list length.** Our first bounds specification asked
322 whether a paper carried at least one zombie citation, which rises mechanically with the number of
323 references. Because the lexical rule needs two markers inside an abstract, it selects long-abstract full
324 articles carrying 44.7 references against 9.2 for the rest. The unconditioned contrast was roughly a
325 hundredfold because the arms differed in length, and the conditioned one is not significant. We now
326 put length on the right-hand side of any per-paper indicator over reference lists.

327 **Debug aggregates merge silently with production shards.** Partial results carry an `<i>of<n>` tag,
328 and a test run left a stray 2130of2446 beside the six production shards. Stage 4 globbed both
329 because the pattern matched either, double-counting that file and shifting the pre-period rate by
330 0.003. The merge now takes only the largest complete sharding.